

Healthcare Clinical Ontology

Knowledge Representation Project Report

Submitted By

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Course: Knowledge Representation

Ontology Language: OWL 2.0

Development Tool: Protégé 5.x

Reasoner: HermiT

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Contents

1	Introduction	2
2	Objectives	2
3	Tools and Technologies	2
4	Ontology Overview	2
5	Class Hierarchy	3
6	Object Properties	4
7	Data Properties	5
8	Individuals	6
9	Property Assertions	7
9.1	Example: Ali	7
9.2	Example: Sara	8
9.3	Data Property Examples	8
10	Reasoning	8
11	DL Queries	9
11.1	Query 1: Patient	9
11.2	Query 2: Doctor	10
11.3	Query 3: Nurse	10
11.4	Query 4: Patients Having Disease	11
12	GitHub Repository	12
13	Conclusion	12
14	References	13

1 Introduction

Healthcare systems generate a large amount of clinical information every day. This information includes patient records, diseases, symptoms, diagnoses, medicines, appointments, treatments, prescriptions, hospitals, doctors, nurses, and insurance details. Managing such information in a structured and meaningful way is important for improving healthcare decision-making and knowledge sharing.

This project presents a **Healthcare Clinical Ontology** developed using **OWL 2.0** in the **Protégé** environment. The ontology represents important healthcare concepts and their semantic relationships. It also supports logical reasoning through the **HermiT Reasoner**, which was used to check the consistency of the ontology.

2 Objectives

The main objectives of this project are:

- To design a healthcare ontology using OWL 2.0.
- To represent healthcare knowledge in a semantic and structured form.
- To define classes, object properties, data properties, and individuals.
- To use domain and range restrictions for meaningful relationships.
- To verify ontology consistency using the HermiT Reasoner.
- To execute DL Queries for retrieving knowledge from the ontology.
- To publish the complete project on GitHub.

3 Tools and Technologies

Tool/Technology	Purpose
Protégé 5.x	Ontology development environment
OWL 2.0	Ontology representation language
HermiT Reasoner	Consistency checking and reasoning
DL Query	Knowledge retrieval
GitHub	Project repository and submission

Table 1: Tools and technologies used in the project

4 Ontology Overview

The developed ontology is named **HealthcareClinicalOntology**. It models a healthcare environment by representing patients, doctors, nurses, hospitals, diseases, medicines, appointments, treatments, prescriptions, medical records, and insurance.

The ontology contains:

- 29 Classes
- 15 Object Properties
- 13 Data Properties
- 42 Individuals
- 267 Axioms

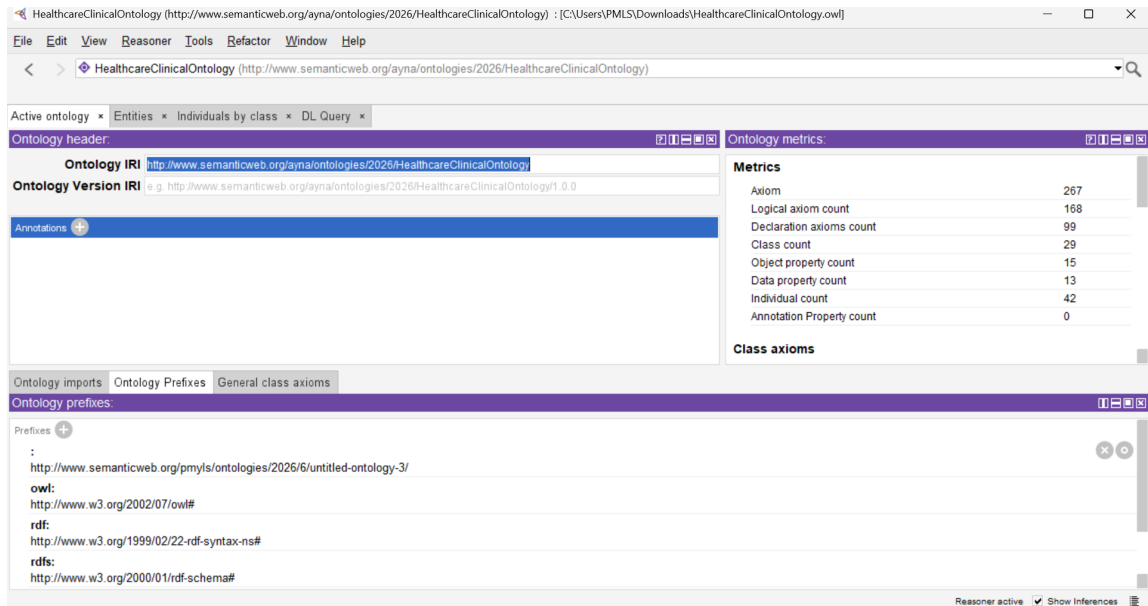


Figure 1: Overview and ontology metrics of the Healthcare Clinical Ontology in Protégé

5 Class Hierarchy

The ontology is organized into several main classes. These classes represent the major concepts of the healthcare domain.

- **Person**
 - Patient
 - Doctor
 - Nurse
- **HealthcareFacility**
 - Hospital
 - Department
 - Laboratory
 - Pharmacy
 - Ward
- **MedicalCondition**
 - Disease

- Symptom
- Diagnosis
- Allergy
- **MedicalService**
 - Appointment
 - Surgery
 - MedicalTest
 - Treatment
 - Vaccination
- **MedicalResource**
 - Medicine
 - Prescription
 - MedicalRecord
 - Equipment
- **AdministrativeEntity**
 - Insurance
 - Billing

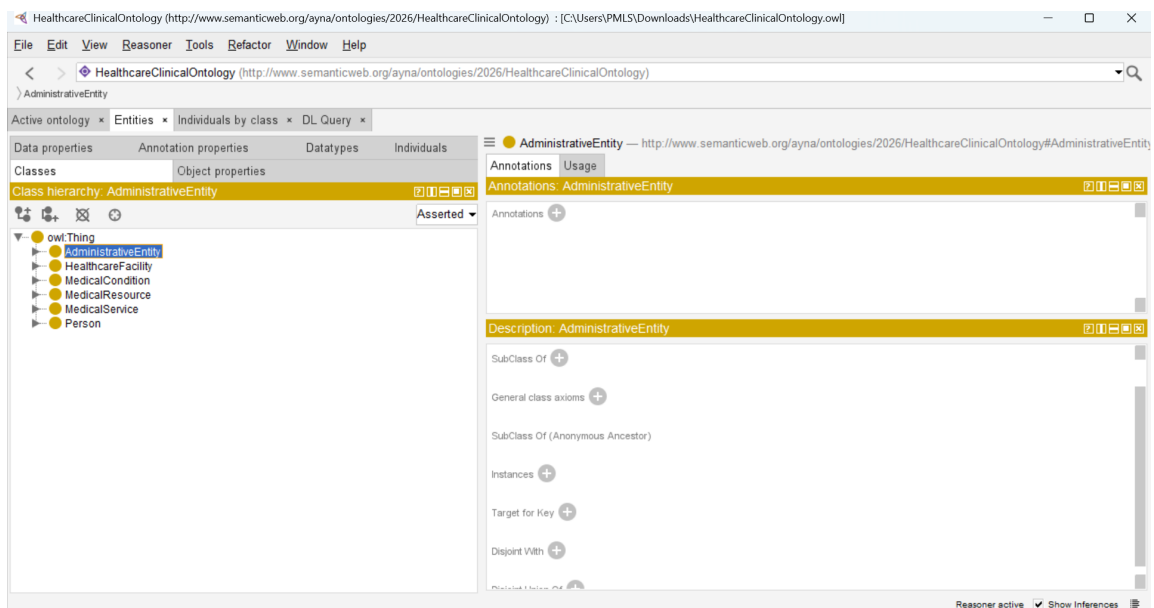


Figure 2: Class hierarchy of the Healthcare Clinical Ontology

6 Object Properties

Object properties define relationships between individuals of different classes.

Object Property	Domain	Range
hasDisease	Patient	Disease
hasSymptom	Patient	Symptom
diagnosedWith	Patient	Diagnosis
treatedBy	Patient	Doctor
prescribedMedicine	Doctor	Medicine
worksIn	Person	Hospital
admittedTo	Patient	Hospital
belongsToDepartment	Person	Department
performsTest	Laboratory	MedicalTest
hasAppointment	Patient	Appointment
hasMedicalRecord	Patient	MedicalRecord
hasInsurance	Patient	Insurance
usesEquipment	Surgery	Equipment
receivesTreatment	Patient	Treatment
hasPrescription	Patient	Prescription

Table 2: Object properties with domain and range

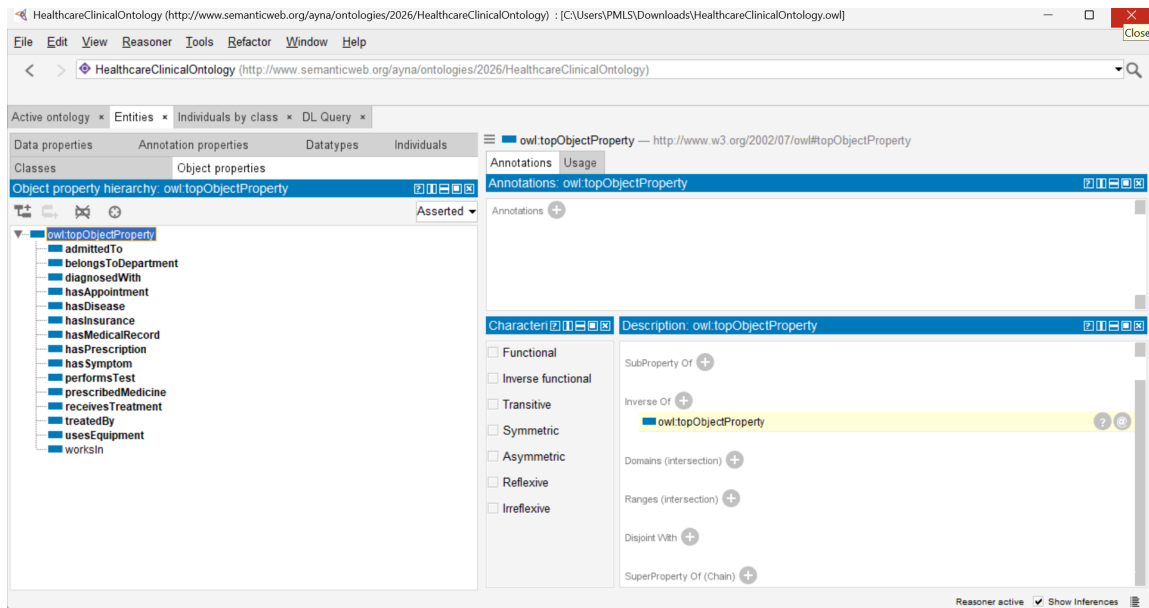


Figure 3: Object properties defined in the ontology

7 Data Properties

Data properties connect individuals with literal values such as names, age, dates, and identification numbers.

Data Property	Domain	Range
patientName	Patient	xsd:string
doctorName	Doctor	xsd:string
nurseName	Nurse	xsd:string
age	Patient	xsd:integer
gender	Patient	xsd:string
bloodGroup	Patient	xsd:string
hospitalName	Hospital	xsd:string
diseaseName	Disease	xsd:string
medicineName	Medicine	xsd:string
appointmentDate	Appointment	xsd:dateTime
diagnosisDate	Diagnosis	xsd:dateTime
insuranceID	Insurance	xsd:string
dosage	Prescription	xsd:string

Table 3: Data properties with domain and range

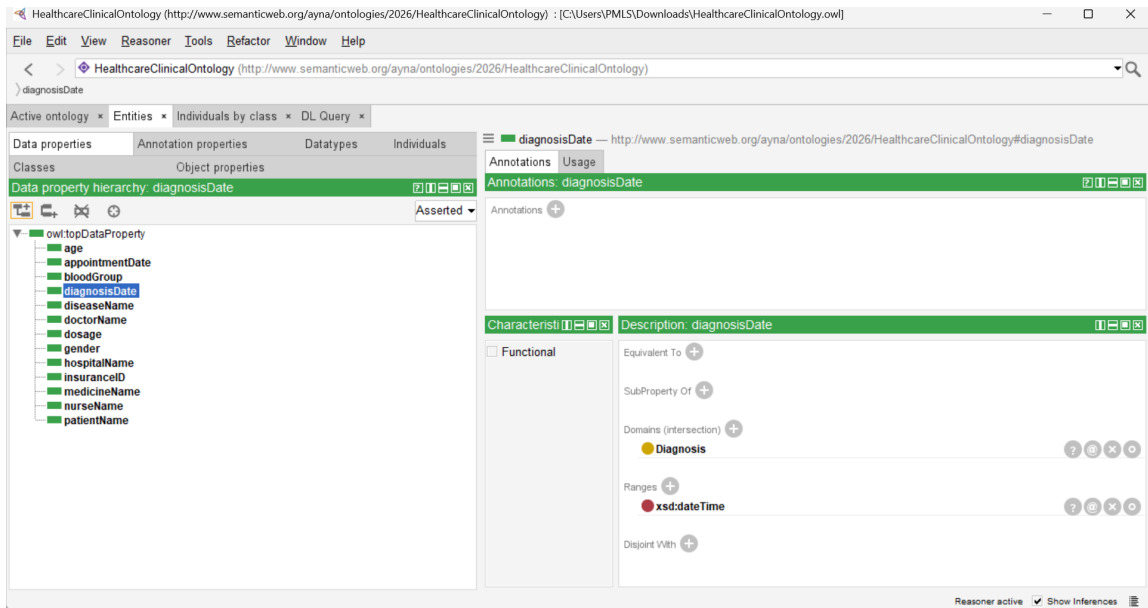


Figure 4: Data properties used in the ontology

8 Individuals

Several individuals were created to demonstrate the ontology.

Class	Individuals
Patient	Ali, Sara
Doctor	DrAhmed, DrFatima
Nurse	NurseAyesha, NurseMaria
Hospital	CityHospital, GeneralHospital
Disease	Diabetes, Flu
Symptom	Fever, Cough
Medicine	Insulin, Paracetamol
Appointment	Appointment001, Appointment002
Diagnosis	Diagnosis001, Diagnosis002
MedicalRecord	Record001, Record002
Insurance	Insurance001, Insurance002
Prescription	Prescription001, Prescription002

Table 4: Main individuals used in the ontology

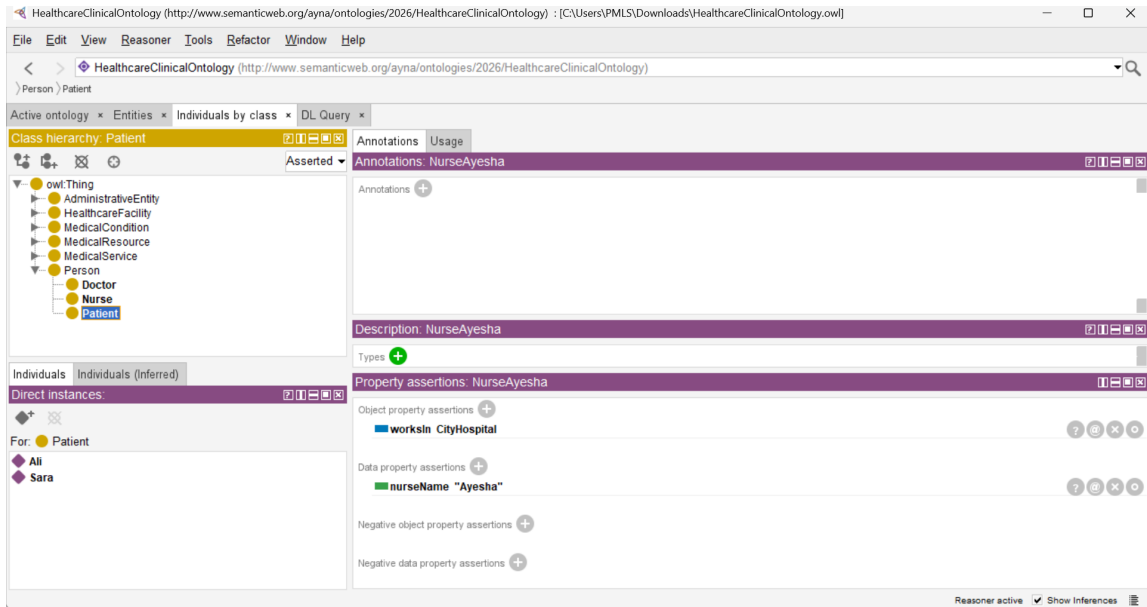


Figure 5: Sample individuals created in the ontology

9 Property Assertions

Property assertions were used to connect individuals.

9.1 Example: Ali

- Ali hasDisease Diabetes
- Ali hasSymptom Fever
- Ali treatedBy DrAhmed
- Ali admittedTo CityHospital

- Ali hasAppointment Appointment001
- Ali hasMedicalRecord Record001
- Ali hasInsurance Insurance001
- Ali receivesTreatment Treatment001
- Ali hasPrescription Prescription001

9.2 Example: Sara

- Sara hasDisease Flu
- Sara hasSymptom Cough
- Sara treatedBy DrFatima
- Sara admittedTo GeneralHospital
- Sara hasAppointment Appointment002
- Sara hasMedicalRecord Record002
- Sara hasInsurance Insurance002
- Sara receivesTreatment Treatment002
- Sara hasPrescription Prescription002

9.3 Data Property Examples

- Ali patientName “Ali”
- Ali age 25
- Sara patientName “Sara”
- Sara age 22
- DrAhmed doctorName “Dr Ahmed”
- DrFatima doctorName “Dr Fatima”
- NurseAyesha nurseName “Ayesha”
- NurseMaria nurseName “Maria”

10 Reasoning

The ontology was checked using the **HermiT Reasoner**. The reasoner verified class hierarchy, object property hierarchy, data property hierarchy, class assertions, object property assertions, and same individuals.

After running the HermiT Reasoner, no logical inconsistency was found. This confirms that the ontology is logically consistent.

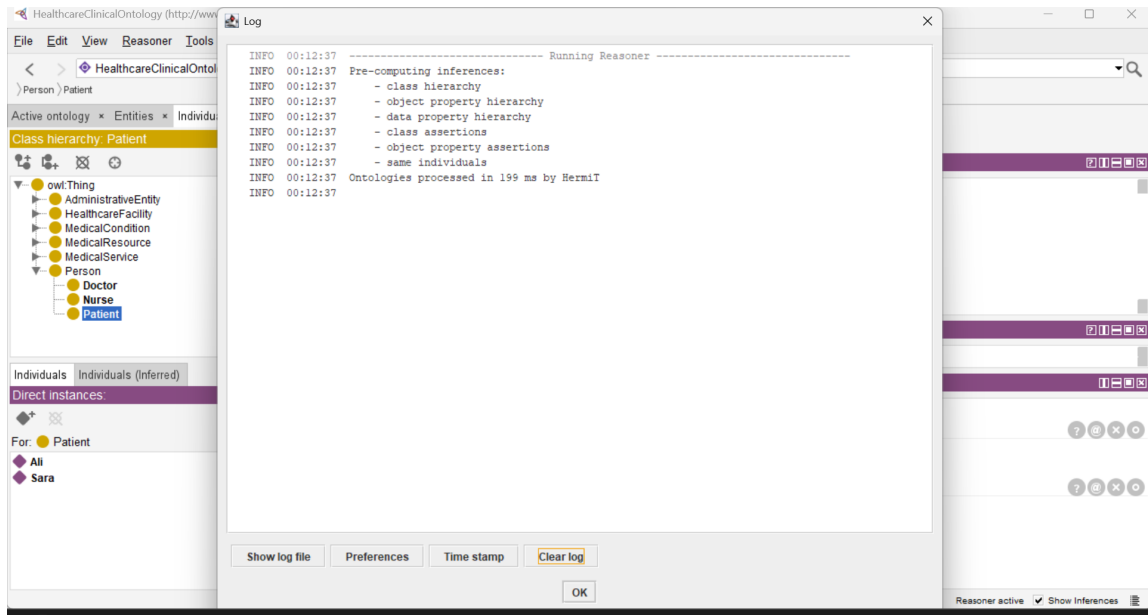


Figure 6: HermiT Reasoner successfully checking ontology consistency

11 DL Queries

DL Queries were used to retrieve information from the ontology.

11.1 Query 1: Patient

Patient

Result:

- Ali
- Sara

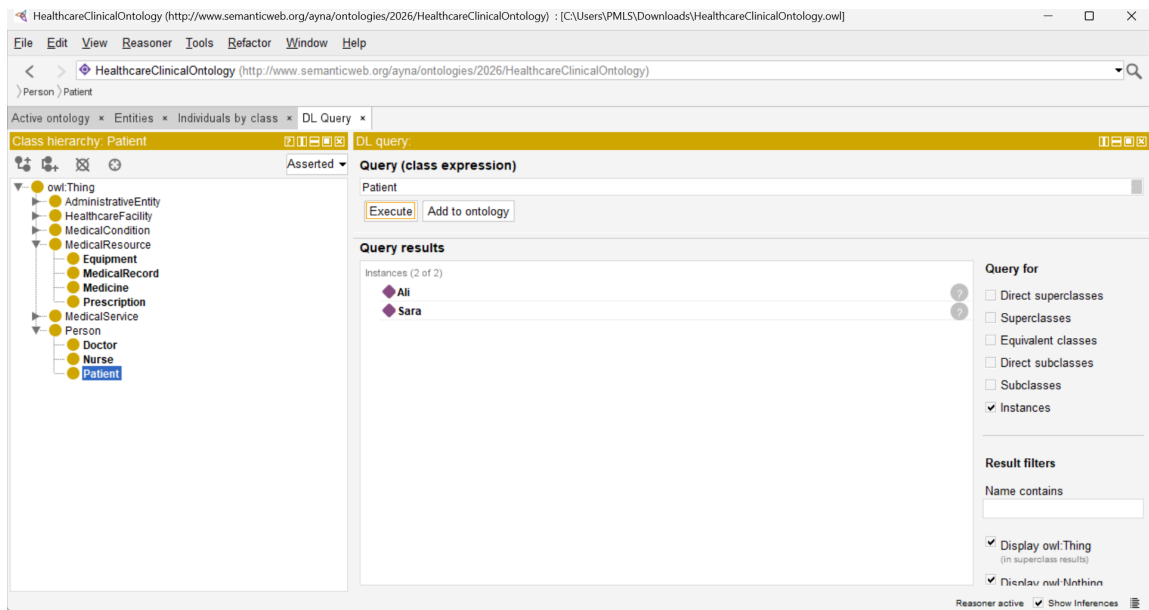


Figure 7: DL Query retrieving all patients

11.2 Query 2: Doctor

Doctor

Result:

- DrAhmed
- DrFatima

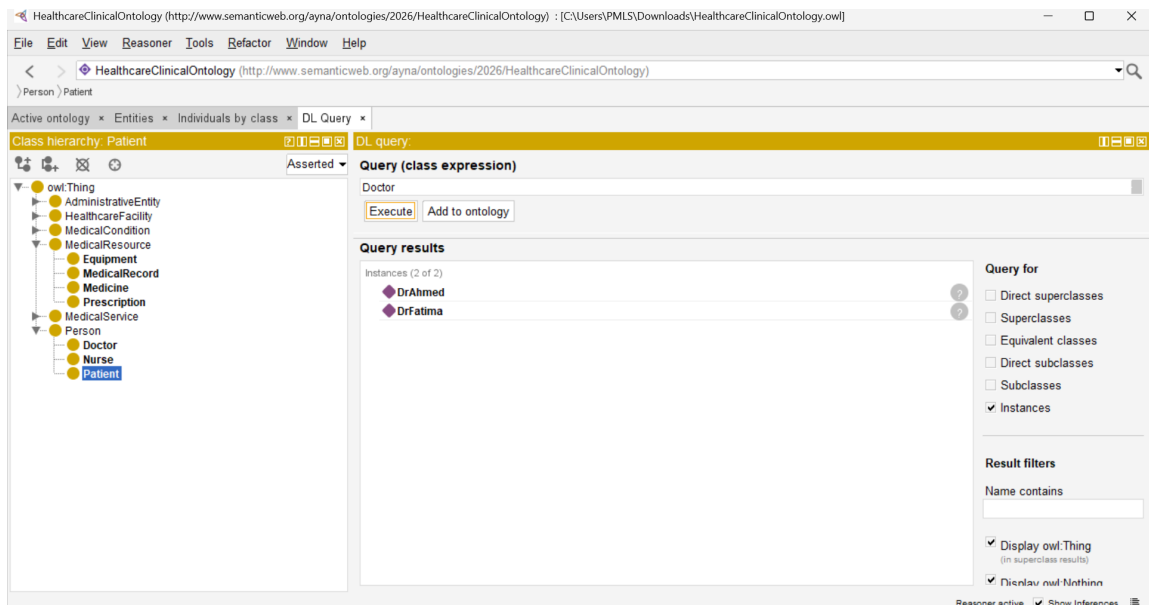


Figure 8: DL Query retrieving all doctors

11.3 Query 3: Nurse

Nurse

Result:

- NurseAyesha
- NurseMaria

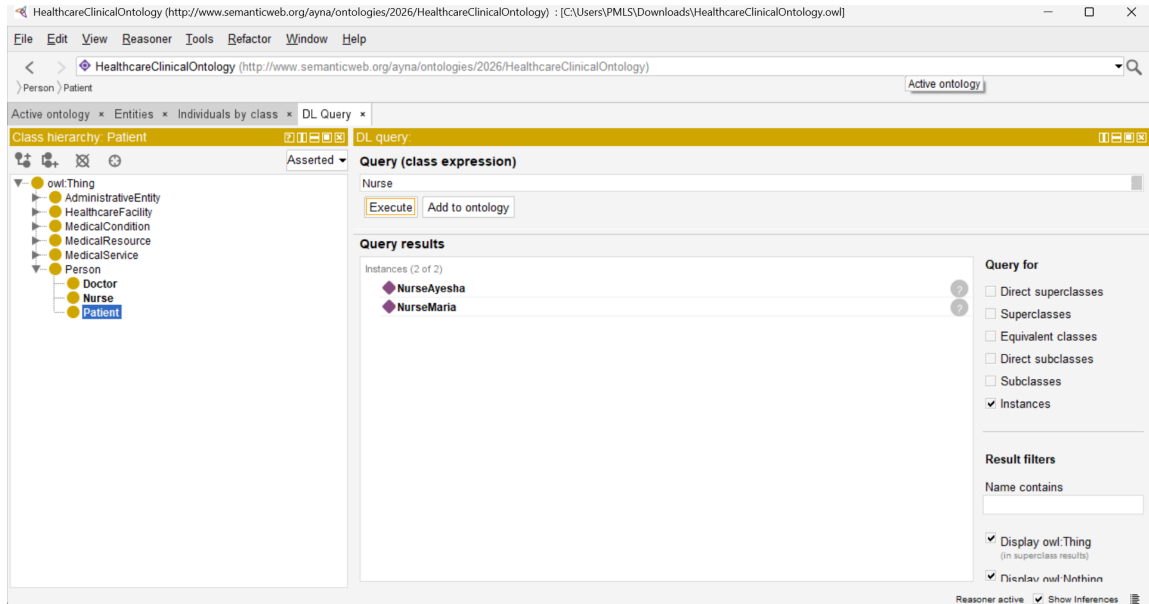


Figure 9: DL Query retrieving all nurses

11.4 Query 4: Patients Having Disease

Patient and hasDisease some Disease

Result:

- Ali
- Sara

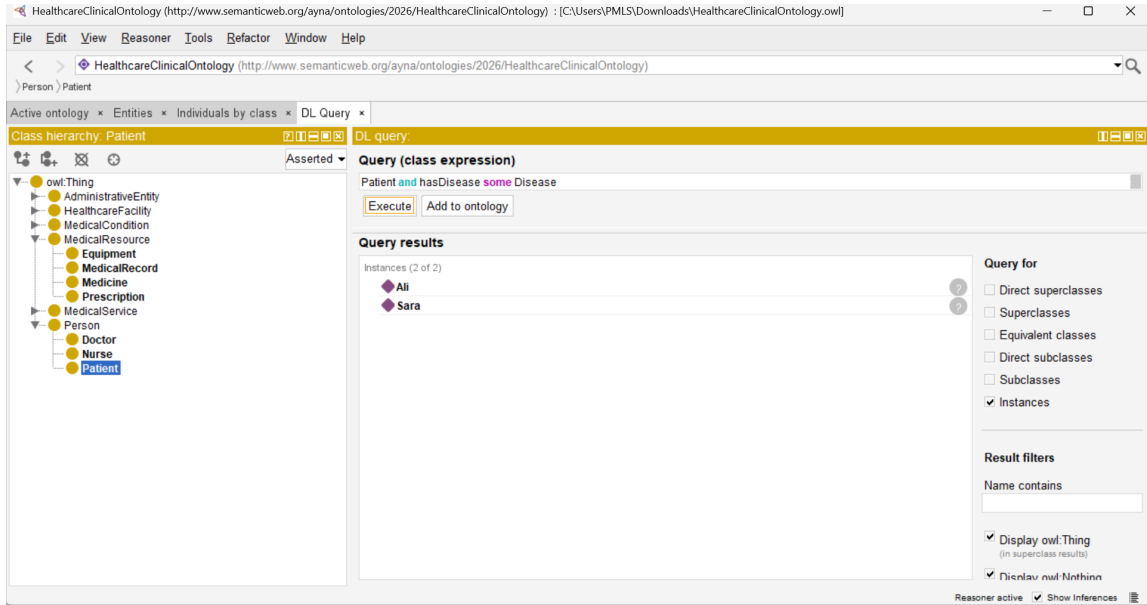


Figure 10: DL Query retrieving patients associated with diseases

12 GitHub Repository

The complete project has been uploaded to GitHub. The repository contains the ontology file, project report, and README documentation.

GitHub Repository:

<https://github.com/aynaramzan/Healthcare-Clinical-Ontology>

13 Conclusion

This project successfully developed a Healthcare Clinical Ontology using OWL 2.0 in Protégé. The ontology represents important healthcare concepts such as patients, doctors, nurses, hospitals, diseases, medicines, appointments, prescriptions, medical records, and insurance.

The ontology includes classes, object properties, data properties, individuals, and property assertions. The HermiT Reasoner confirmed that the ontology is logically consistent. DL Queries were executed successfully to retrieve relevant healthcare knowledge. This project demonstrates how ontology engineering can support semantic knowledge representation in the healthcare domain.

14 References

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3. HermiT OWL Reasoner Documentation.
4. Gruber, T. R. (1993). A Translation Approach to Portable Ontology Specifications.
5. Noy, N. F., and McGuinness, D. L. (2001). Ontology Development 101: A Guide to Creating Your First Ontology.